Gianmarco Fabbri

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Profile

Cybersecurity Engineering M.Sc. student at Politecnico di Torino, focused on cloud and Kubernetes security: runtime detection, automated incident response and hardening, with a growing interest in the security of AI and LLM systems.

Education

Politecnico di Torino Oct 2025 – Expected Oct 2027
M.Sc. in Cybersecurity Engineering *Turin, Italy*

- Coursework: Cryptography, Web Applications. In progress: AI and Cybersecurity; Data Protection, Privacy and Anonymity; Networks & Cloud Technologies and Security; Information Systems Security; Security Verification and Testing; Threat Intelligence and Incident Management.

Alma Mater Studiorum – University of Bologna Sep 2022 – Oct 2025
B.Sc. in Computer Science and Engineering *Forlì-Cesena, Italy*

- Thesis: *The Post-Quantum Era and Lattice-Based Cryptography*. RSA and ECC under quantum attacks (Shor, Grover) and the lattice-based schemes CRYSTALS-Kyber and CRYSTALS-Dilithium (NIST **ML-KEM** and **ML-DSA**).

Experience

Incident Response & Site Recovery (pro bono) – Law firm 2026
Compromised WordPress website: Linux hosting, MySQL, Python, Docker *Forlì, Italy*

- Investigated an SEO-spam compromise: found backdoors in the WordPress core files and a malicious payload in the database, and correlated about **170,000** log events in Python.
- Rebuilt the site from official sources and tested it in Docker, bringing it back online in **5 days** while coordinating with the hosting provider; hardened it with a WAF, two-factor authentication and credential rotation.

Software Engineer Intern – myDev s.r.l. (VEM Group) Feb 2025 – Mar 2025
Curricular internship: Cisco Contact Center integration (C#, ASP.NET Core, Blazor) *Forlì, Italy*

- Team of 2 with a company software engineer: web app that lets a client manage its Cisco Contact Center configuration without the native Cisco tools; Repository–Service backend whose APIs generate the XML files.
- Analysed the client's system and requirements, validated the app with end users and delivered ahead of schedule.

Projects

Kubernetes Automated Incident Response | *Python, Kubernetes, Falco, Cilium* GitHub | 2026

- Falco (eBPF) detects attacks on a deliberately vulnerable pod (reverse shell, ptrace anti-debugging, network tools via a custom rule); the pod is isolated by a Cilium policy, kept for investigation and replaced.
- Python response handler behind an authenticated webhook, scoped RBAC, non-root container; **22 unit tests** in GitHub Actions CI and an end-to-end test that replays the attacks on a kind cluster.

PTL – Periodic Task Layer for FreeRTOS | *C, FreeRTOS, ARM Cortex-M3* GitHub | 2026

- Team of 4. Real-time middleware that adds deadline-aware periodic tasks to FreeRTOS on ARM Cortex-M3 (QEMU), with three overrun policies (SKIP, KILL, CATCH_UP) and automated regression tests.

Theater Booking System | *React, Node.js / Express, SQLite* GitHub | 2026

- Individual project: Passport.js sessions, scrypt password hashing, roles, admin TOTP, transactional booking checks.

Other projects: Videogame Planet Shop (MySQL, team of 2) | 100 Days War (Java, MVC, team of 4) 2024 – 2025

Technical Skills

Security: Incident response, log analysis, WAF, 2FA / TOTP, runtime detection (Falco), network isolation (Cilium), RBAC
Cloud & DevOps: Kubernetes, Helm, kind, Docker, eBPF, GitHub Actions CI, Linux, Bash
Cryptography: RSA, ECC, digital signatures, hash functions, post-quantum cryptography (ML-KEM, ML-DSA)
Programming: Python, C, C#/.NET (ASP.NET Core, Blazor), Java, JavaScript (React, Node.js), PHP, SQL
Languages: Italian (native), English (B2, Cambridge B2 First), French (A2)